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Forensica AI — Hackathon Documentation Landing Page

Overview

Forensica AI is an explainable AI–powered document verification platform that detects forged or manipulated documents. Visitors should review the prototype/demo and download the verification report or project brief for judging evaluation.

Conversion Goal

- **Primary CTA:** "View Demo & Report" (opens interactive prototype and sample report)
- **Secondary CTA:** "Download Project Brief" (PDF of system design and evaluation)

Page Shell

- **Layout:** Single page, vertical scroll, clearly-separated sections matching judge evaluation checklist
- **Theme:** Professional, high-contrast (dark slate background #0F172A, accent teal #06B6D4), readable sans-serif, ample whitespace, code/plain-text style for technical blocks

Sections

Hero

- **Headline:** Explainable AI for Document Forgery Detection
- **Subheadline:** Forensica AI flags manipulated PDFs, certificates, IDs and images — then explains why with visual overlays and a structured verification report.
- **CTA:** "View Demo & Report"
- **Visual:** Split-screen: left — sample document with highlighted suspicious regions; right — snapshot of verification report (risk score, anomaly list)

Problem/Pain Point

- **Content:**
  - Document fraud affects college admissions, scholarship checks, job applications, and government processes.
  - Manual checks are slow, inconsistent, and require specialists.
  - Organizations need fast, explainable automation to reduce risk and speed decisions.
- **Tone:** Empathetic, urgent, practical

Describe what you'd like to add, update, or fix in the spec...

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with confidence scores — aligning directly with Track C's Explainable AI for Document Forgery Detection.

- **Visual:** Flow graphic showing Upload → Analysis → Visual Overlays → Report

System Architecture

- **Content:** Three components:
  1. Frontend Application — web UI for uploads, progress, report review.
  2. Backend API — FastAPI handles ingestion, job orchestration, authentication.
  3. Forensic Analysis Pipeline — modular stages (preprocess, OCR, pixel forensics, layout/font checks, metadata, scoring).
- **Integration Note:** Frontend polls backend job endpoints and retrieves JSON report + overlay assets for client rendering.

How the System Works

1. The user uploads a document (PDF, JPG, PNG).
2. The backend securely stores the encrypted document.
3. The document enters the forensic analysis pipeline.
4. Multiple modules analyze the file including OCR extraction, pixel anomaly detection, layout analysis, font consistency checks, and metadata inspection.
5. Each module produces a score and potential anomaly signals.
6. A risk scoring engine aggregates the results to compute the final authenticity score.
7. The frontend renders visual overlays highlighting suspicious regions.
8. A structured verification report is generated explaining the results.

Technical Pipeline (Concise)

- Document Preprocessing: render PDF pages, normalize images.
- OCR Extraction: Tesseract-based multilingual text extraction and structural parsing.
- Pixel Forensics: compression/noise inconsistency and editing artifact detection via OpenCV.
- Layout Consistency: margins, spacing, and alignment checks.
- Font Consistency: detect mixed or suspicious typography.
- Metadata Forensics: EXIF and file metadata analysis.
- Risk Scoring Engine: weighted aggregation into final authenticity score.

Explainable AI Output

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- CV & Analysis: OpenCV, NumPy, Pillow, PyMuPDF, Tesseract OCR
- Security & Storage: Firebase Auth, Firestore, Firebase Cloud Storage
- Deployment: Docker, Netlify (frontend), Render or similar for backend

Storage and Security

- **Model:** Files encrypted at rest; documents tied to user UID; access requires verified auth tokens. Role-based access for judges/demonstrators. Audit log for document access and analysis runs.

Deployment Strategy

- **Hybrid:** Local-first for demos and development; supports cloud deployment with Dockerized backend and Netlify frontend. Firebase used for auth/storage in cloud mode to scale securely.

Key Features

- AI-powered forgery detection
- Explainable anomaly reports with overlays
- Automated verification report generation
- Encrypted storage and user-scoped data isolation
- Batch processing and document history tracking

Future Research Directions

- Deep-learning forgery models (transformer/CNN ensembles)
- Expanded regional language OCR models
- Optional blockchain anchoring for tamper-evident verification
- Enterprise-scale ingestion and role-based workflows

Social Proof / Judging Signals

- Prototype demo + sample report available
- Example metric: "Detected 95% of synthetic edits in internal test set (precision 92%)" — sample test results included in downloadable brief
- Team contact and repo link for evaluators

Final CTA

- **Content:** Evaluate the interactive demo and sample report to judge explainability and detection quality.
- **CTA:** "View Demo & Report"

Design Notes

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Returns a structured verification report including risk score and detected anomalies.

- GET /preview/{document_id}

Returns a document preview used for overlay visualization.

- GET /health

Returns backend health status.

Project Repository

GitHub Repository

https://github.com/NovocaineX/vortex-hackathon26

The repository contains the full source code including:

- FastAPI backend
- forensic analysis pipeline
- frontend single-page application
- deployment configuration

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